Ch 6.2: Shrinkage - The Lasso

Lecture 18 - CMSE 381

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Announcements

Last time:

- Ridge Regression

This time:

- The Lasso

Announcements:

- HW4 due Sunday
- HW5 released
- Think about the project, choose partner in crime

CMSE381_F2025_Schedule : Schedule

	M	9/22	***Project Day & Review***		
	W	9/24	**Midterm #1**		
12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso

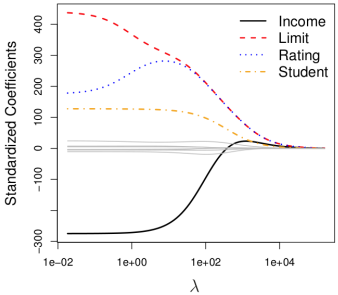
Ridge regression

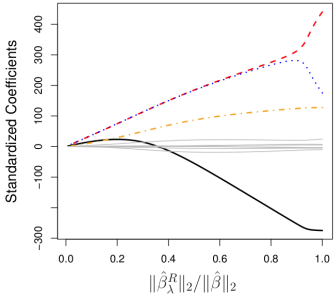
Before:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

After:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$





Scale equivariance (or lack thereof)

Scale equivariant: Multiplying a variable by c (cX_i) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

- Least squares is scale equivariant
- Ridge regression is not

Solution: standardize predictors

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

- Ex: `income variable`. If you put in income by \$1,000's of dollars, the answer you would get is quite different. Might emphasize a different coefficient.
- Least squares is scale equivariant
- Ridge regression very much is not
- $X_j\hat{\beta}^R_{j,\lambda}$ depends not only on λ but also on values of other predictors

Section 2

The Lasso

Same goal as before

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0 *Ridge shrunk but DID NOT GET RID OF which is what we actually want*

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso: Least Absolute Shrinkage and Selection Operator

The lasso

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

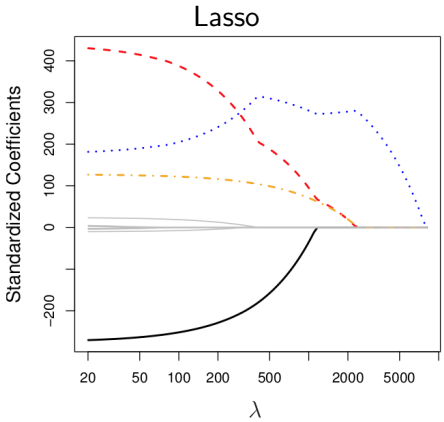
- ℓ_1 penalty instead of ℓ_2 penalty
- ℓ_1 norm: $\|\beta\|_1 = \sum|\beta_j|$
- ℓ_2 norm: $\|\beta\|_2 = \sum\beta_j^2$
- ℓ_p norm: $\|\beta\|_p = (\sum|\beta_j|^p)^{1/p}$
- ℓ_∞ norm: $\|\beta\|_\infty = \max\{|\beta_i|\}$

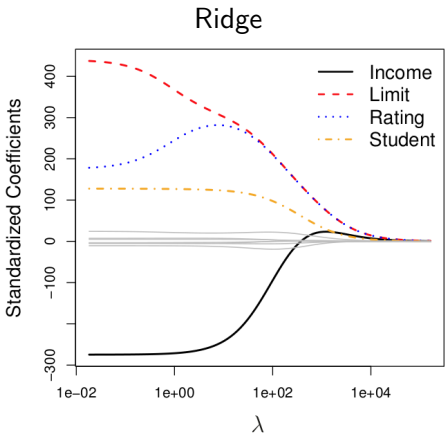
Subsets with lasso

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

- Result of ℓ_1 penalty is that it forces some coefficients to be exactly 0
- Can be interpreted as something more like subset selection, but here called *variable selection*
- Yield are *sparse* models, that is those using only a subset of variables
- Selecing good λ is critical

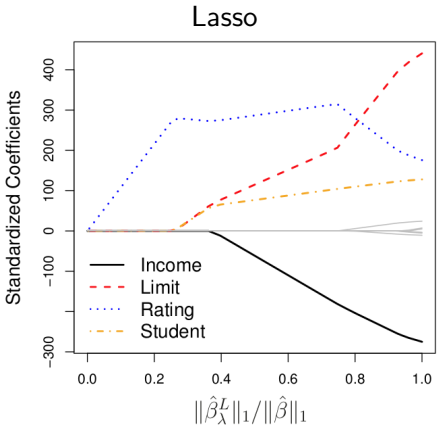
An example on `Credit` data set

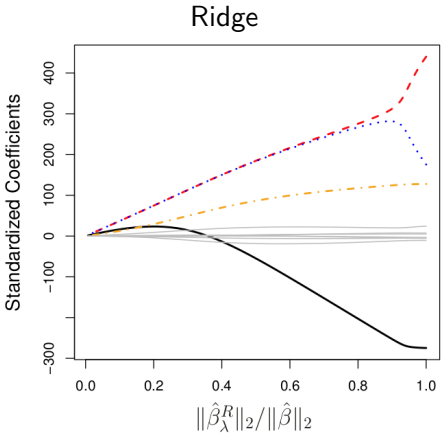




- $\lambda = 0$: Both are least squares
- $\lambda = \infty$: Everything is 0
- only lasso has 0 efficients at finite λ

More example on `Credit` data set





- Note that ridge always includes all variables, even if small coefficient
- Lasso produces different sized subsets depending

Why the hell lasso can select variable (while ridge cannot)?

Alternative formulation of lasso & ridge regression (play more with ℓ_p)

$$\min_{\beta} \sum (y_i - \hat{y}_i)^2 \text{ where } \sum |\beta_j| \leq s$$

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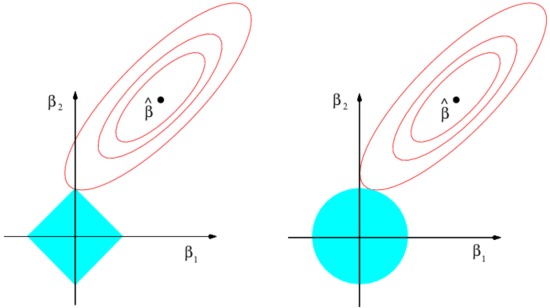
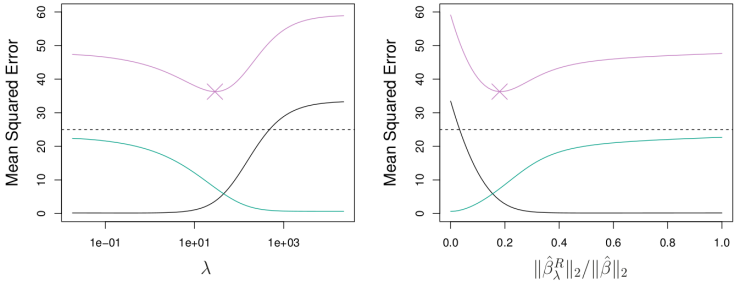


Bias-Variance tradeoff



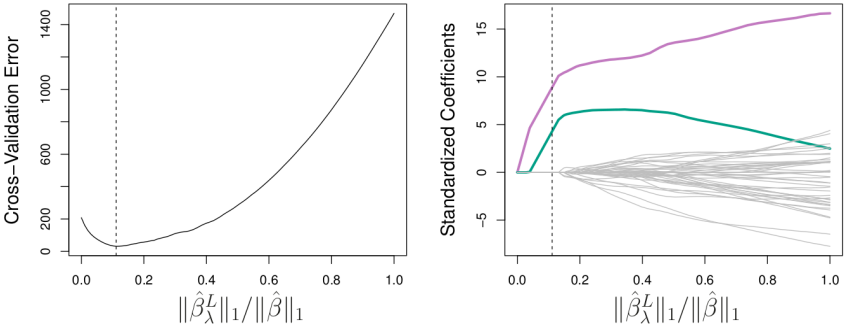
Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Horizontal dashed line is minimum possible test MSE

Using Cross-Validation to find λ

- Choose a grid of λ values
- Compute the (k-fold) cross-validation error for each value of λ
- Select the tuning parameter value λ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

10-fold CV choice of λ for lasso and simulated data



- Data generated with $p = 45$ variables, $n = 50$ data points, only 2 variables actually involved
- The vertical dashed line: choice of λ
- Right: Two colored lines are predictors related to the response (signal), grey are unrelated (noise)
- Correctly gives much larger coefficient estimates to the two signal predictors, but also the min CV error corresponds to a set of coefficient estimates for which only the signal variables are non-zero.
- Note that least squares solution doesn't even emphasize one of the variables (green one)

Coding example

Ridge vs Lasso

Ridge Regression:

- Coefficients are shrunk towards 0, but not forced to be exactly 0
- Tends to be better than lasso when many predictors have significant effect on output
- More robust to outliers

Lasso:

- Coefficients are shrunk towards 0, and some are forced to be exactly 0
- Tends to be more interpretable since we can see which variables are non-zero (rather than having to decide how small is small enough)
- More sensitive to outliers

TL;DR - Original forumlation

Least Squares:

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

Ridge:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| = RSS + \lambda \sum_{j=1}^{p} |\beta_j|$$

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